



Review

84%**16**

Out of 19 points

2:45

Avg Time Per Question

Your Answers:

Vignette

Quentin Abay, CFA, is an analyst for a private equity firm interested in purchasing Bickchip Enterprises, a conglomerate. His first task is to determine the trends in ROE and the main drivers of the trends using DuPont analysis. To do so he gathers the data in Exhibit 1.

Exhibit 1

Selected Financial Data for Bickchip Enterprises (€ Thousands)

	2020	2019	2018
Revenue	72,448	66,487	55,781
Earnings before interest and tax	6,270	4,710	3,609
Earnings before tax	5,101	4,114	3,168
Net income	4,038	3,345	2,576
Asset turnover	0.79	0.76	0.68
Assets/Equity	3.09	3.38	3.43

After conducting the DuPont analysis, Abay believes that his firm could increase the ROE without operational changes. Further, Abay thinks that ROE could improve if the company divested segments that were generating the lowest returns on capital employed (total assets less non-interest-bearing liabilities). Segment EBIT margins in 2020 were 11 percent for Automation Equipment, 5 percent for Power and Industrial, and 8 percent for Medical Equipment. Other relevant segment information is presented in Exhibit 2.

Exhibit 2

Segment Data for Bickchip Enterprises (€ Thousands)

Operating Segments	Capital Employed			Capital Expenditures (Excluding Acquisitions)		
	2020	2019	2018	2020	2019	2018
Automation Equipment	10,705	6,384	5,647	700	743	616
Power and Industrial	15,805	13,195	12,100	900	849	634
Medical Equipment	22,870	22,985	22,587	908	824	749
	49,380	42,564	40,334	2,508	2,416	1,999

Abay is also concerned with earnings quality, so he intends to calculate Bickchip's cash-flow-based accruals ratio and the ratio of operating cash flow before interest and taxes to operating income. To do so, he prepares the information in Exhibit 3.

Exhibit 3

Earnings Quality Data for Bickchip Enterprises (€ Thousands)

	2020	2019	2018
Net income	4,038	3,345	2,576
Net cash flow provided by (used in) operating activity ^a	9,822	5,003	3,198
Net cash flow provided by (used in) investing activity	(10,068)	(4,315)	(5,052)
Net cash flow provided by (used in) financing activity ^b	(5,792)	1,540	(2,241)
Average net operating assets	43,192	45,373	40,421
^a includes cash paid for taxes of:	(1,930)	(1,191)	(1,093)
^b includes cash paid for interest of:	(1,169)	(596)	(441)

1 Multiple Choice 1 / 1 point

Over the three-year period presented in Exhibit 1, Bickchip's return on equity is *best* described as:

- ☐ A. stable.
- ☐ B. trending lower.
- ☒ C. trending higher.

Feedback

C is correct. The ROE has been trending higher. ROE can be calculated by multiplying (net profit margin) $\times$ (asset turnover) $\times$ (financial leverage). Net profit margin is net income/sales. In 2018 the net profit margin was $2,576/55,781 = 4.6\%$ and the ROE = $4.6\% \times 0.68 \times 3.43 = 10.8\%$. Using the same method, ROE was 12.9 percent in 2019 and 13.6 percent in 2020.

2 Multiple Choice 1 / 1 point

Based on the DuPont analysis, Abay's belief regarding ROE is *most likely* based on:

- ☒ A. leverage.
- ☐ B. profit margins.
- ☐ C. asset turnover.

Feedback

A is correct. The DuPont analysis shows that profit margins and asset turnover have both increased over the last three years, but leverage has declined. The reduction in leverage offsets a portion of the improvement in profitability and turnover. Thus, ROE would have been higher if leverage had not decreased.

3 Multiple Choice 1 / 1 point

Based on Abay's criteria, the business segment *best* suited for divestiture is:

- ☐ A. medical equipment.
- ☒ B. power and industrial.
- ☐ C. automation equipment.

Feedback

B is correct. The Power and Industrial segment has the lowest EBIT margins but uses about 31 percent of the capital employed. Further, Power and Industrial's proportion of the capital expenditures has increased from 32 percent to 36 percent over the three years. Its capital intensity only looked to get worse, as the segment's percentage of total capital expenditures was higher

than its percentage of total capital in each of the three years. If Abay is considering divesting segments that do not earn sufficient returns on capital employed, this segment is most suitable.

4 Multiple Choice 1 / 1 point

Bickchip's cash-flow-based accruals ratio in 2020 is *closest* to:



☒ A.
9.9%.

☐ B. 13.4%.

☐ C. 23.3%.

Feedback

A is correct. The cash-flow-based accruals ratio = $[NI - (CFO + CFI)] / (\text{Average NOA}) = [4,038 - (9,822 - 10,068)] / 43,192 = 9.9\%$.

5 Multiple Choice 0 / 1 point

The cash-flow-based accruals ratios from 2018 to 2020 indicate:

☐ A. improving earnings quality.



☒ B. deteriorating earnings quality.

Correct Answer: **A. improving earnings quality.**

☐ C. no change in earnings quality.

Feedback

A is correct. The cash-flow-based accruals ratio falls from 11.0 percent in 2018 to 5.9 percent in 2019, and then rises to 9.9 percent in 2020. However, the change over the three-year period is a net modest decline, indicating a slight improvement in earnings quality.

6 Multiple Choice 0 / 1 point

The ratio of operating cash flow before interest and taxes to operating income for Bickchip for 2020 is *closest* to:

☐ A. 1.6.

☐ B. 1.9.



☒ C.
2.1.

Correct Answer: **B. 1.9.**

Feedback

B is correct. Net cash flow provided by (used in) operating activity has to be adjusted for interest and taxes, as necessary, in order to be comparable to operating income (EBIT). Bickchip, reporting under IFRS, chose to classify interest expense as a financing cash flow so the only necessary adjustment is for taxes. The operating cash flow before interest and taxes = $9,822 + 1,930 = 11,752$. Dividing this by EBIT of 6,270 yields 1.9.

7 Multiple Choice 1 / 1 point

Based on the ratios for operating cash flow before interest and taxes to operating income, Abay should conclude that:



☒ A. Bickchip's earnings are backed by cash flow.

☐ B. Bickchip's earnings are not backed by cash flow.

☐ C. Abay can draw no conclusion due to the changes in the ratios over time.

Feedback

A is correct. Operating cash flow before interest and taxes to operating income rises steadily (not erratically) from 1.2 to 1.3 to 1.9. The ratios over 1.0 and the trend indicate that earnings are supported by cash flow.

Vignette

Ione Bailey is a financial analyst who is examining the most recent annual report of Eastview Inc., a US-based company that complies with US GAAP. Bailey wants to determine Eastview's ROE as well as the trend in its ROE components by using DuPont analysis. Data used for this purpose is presented in Exhibit 1.

Exhibit 1

	Most Recent Year	Previous Year	Two Years Ago
Revenue (USD million)	750	680	600
Net income (USD million)	60	55	50
Total liabilities (USD million)	230	220	200
Total asset turnover ratio	1.30	1.25	1.20

Bailey also wants to evaluate Eastview's three geographic operating segments to assess their ROI and growth during the most recent year. For this purpose, she prepares the data presented in Exhibit 2 for the most recent year.

Exhibit 2

	Europe	Asia	Americas
Segment EBIT margin %	12.5	13.1	12.8
Segment % of total assets	26.5	31.2	42.3
Segment % of total capital expenditures	32.2	29.1	38.7

Next, while reading the notes to the financial statements in the annual report, Bailey observes that Eastview's property, plant and equipment have depreciable lives that are 20% longer than its industry competitors' average. For comparison purposes, Bailey adjusts the financial statements to determine the effect of the difference on the company's financial ratios and information.

Finally, in order to assess Eastview's accruals and earnings quality, Bailey calculates the accruals ratios that are presented in Exhibit 3.

Exhibit 3

	Most Recent Year	Previous Year	Two Years Ago
Balance-sheet-based accruals ratio (%)	7.9	3.8	3.1
Cash-flow-based accruals ratio (%)	4.7	4.4	2.6

8 Multiple Choice 1 / 1 point

Based on the 3 years of data provided for DuPont analysis, Eastview's financial leverage has been:

☐ trending lower.

Incorrect Answer Feedback: Incorrect because the 3-year trend has been stable.

☒ stable.

Correct Answer Feedback: Correct because financial leverage can be determined from the information provided as follows:

	Formula	Most recent year	Previous year	Two years ago
Total assets	Revenues/Total asset turnover	$750/1.30 = 576.92$	$680/1.25 = 544$	$600/1.2 = 500$
Total shareholders equity	Total assets – Total liabilities	$576.92 - 230 = 346.92$	$544 - 220 = 324$	$500 - 200 = 300$
Financial leverage	Total assets/Total shareholders' equity	$576.92/346.92 = 1.663$	$544/324 = 1.679$	$500/300 = 1.667$

Accordingly, the 3-year trend in financial leverage has been stable at 1.7.

☐ trending higher.

Incorrect Answer Feedback: Incorrect because the 3-year trend has been stable.

9 Multiple Choice 1 / 1 point

Based on the segment information, Eastview is growing its business the *most* in:

☐ Asia.

Incorrect Answer Feedback: Incorrect because Europe is growing fastest. In the most recent year, the Asia segment has the highest EBIT margin % but growth should be measured using capital expenditures % to total asset %.

☒ Europe.

Correct Answer Feedback: Correct because the ratio of the total capital expenditures % to the total assets % measures segment business growth. A ratio of *less than* 1 indicates that the segment is being allocated a lesser proportion of capital expenditures than its proportion of total assets; if a trend develops, the segment will become less significant over time. A ratio of greater than 1 indicates the company is growing the segment; the segment is receiving a 'growth allocation' of capital spending.



Accordingly, the segment growth evaluation is as follows:

Europe: $32.2\% \div 26.5\% = 1.215$

Asia: $29.1\% \div 31.2\% = 0.933$

Americas: $38.7\% \div 42.3\% = 0.915$

Accordingly, Europe is the geographic segment that is growing fastest.

☐ Americas.

Incorrect Answer Feedback: Incorrect because Europe is growing fastest. The Americas segment has the highest total capital expenditure % as well as the highest total assets % but growth should be measured using capital expenditures % to total asset %.

10 Multiple Choice 1 / 1 point

After Bailey makes the adjustment for Eastview's treatment of its property, plant and equipment, which of the following will increase?

☒ Debt-to-equity ratio


Correct Answer Feedback: Correct because the debt-to-equity ratio (Total debt ÷ Total shareholders' equity) will increase. Net income and therefore shareholders' equity will decrease after Bailey makes the adjustment for a greater amount of depreciation expense. There is no effect on total debt.

☐ Book value per share

Incorrect Answer Feedback: Incorrect because book value per share (Common shareholders' equity ÷ Total number of common shares outstanding) will decrease. Net income and therefore shareholders' equity will decrease after Bailey makes the adjustment for a greater amount of depreciation expense. There is no effect on the total number of common shares outstanding.

☐ Operating profit margin

Incorrect Answer Feedback: Incorrect because the operating profit margin (Operating profit ÷ Total revenues) will decrease after Bailey makes the adjustment to record a greater amount of depreciation expense to make Eastview more comparable with its competitors. Operating profit will be reduced. There is no effect on total revenues.

11 Multiple Choice 1 / 1 point

Based on the Bailey's calculations, Eastview's balance-sheet-based accruals ratio is:

☐ trending lower and improving.

Incorrect Answer Feedback: Incorrect because the balance-sheet-based accruals ratio is deteriorating and trending higher.

☐ trending lower and deteriorating.

Incorrect Answer Feedback: Incorrect because the balance-sheet-based accruals ratio is trending higher.

☒ trending higher and deteriorating.


Correct Answer Feedback: Correct because the percentage trend of the balance-sheet-based accruals ratio is increasing over the 3-year period. This trend indicates that accruals are an increasing factor in the financial results.

Vignette

Natalyn Winslow is an analyst working for High Impact Investments (HII), which manages a variety of investment funds. She analyzes the recent performance of Dorado Beverages SA (DB), a multinational brewer of carbonated beverages that recently entered the energy drinks market. As a core holding, DB's prominence in Europe provides geographic diversification for HII which tends to be North-America-centric.

Winslow refers to DB's most recent year and previous year balance sheets and comments: "DB is continuing to follow a growth-by-acquisition strategy." In addition, Winslow adjusts DB's most recent year income statement to determine the company's net income under the direct control of management by using information (in EUR millions) in Exhibit 1.

Exhibit 1

	Most Recent Year
Income from operations	1,765
Income from investments in associates	191
Interest expense	(648)
Taxes	(238)
Net income	1,070
Of which:	
Net income attributable to non-controlling interests	86
Net income attributable to shareholders	984

Winslow and her team assess the quality of DB's overall performance. The consensus is positive, but Winslow disagrees with her team for the following reasons:

Reason 1: The rate of fluctuation in the cash-flow-based accruals ratio is low.

Reason 2: Balance-sheet-based accruals are trending down relative to total assets.

Reason 3: Income from operations is consistently higher than cash generated from operations.

Winslow and her team notice from DB's annual report that the company has four segments. They analyze DB's segments by using information (in EUR millions) in Exhibit 2.

Exhibit 2

	Most Recent Year			Previous Year
	Sales	EBIT	Assets	Sales
Carbonated drinks - Europe	2,845	955	12,416	2,758
Carbonated drinks - North America	1,913	558	5,876	1,721
Carbonated drinks - Asia Pacific	535	172	2,661	503
Energy drinks	284	16	1,219	202
Total	5,577	1,701	22,172	5,184

After performing the segment analysis, Winslow and her team conclude on the following statements:

- Statement 1: The sales growth rate in carbonated drinks - North America is stronger than that of energy drinks.
- Statement 2: Carbonated drinks - Europe's EBIT margin is superior to that of carbonated drinks - North America.
- Statement 3: Asset turnover in carbonated drinks - Europe is better than that of DB overall.

12 Multiple Choice 1 / 1 point

Which of the following balance sheet line items is *most* appropriate to justify Winslow's comment about DB's growth strategy?

☒ Goodwill

Correct Answer Feedback: Correct because goodwill arises when a company acquires another business.

Goodwill and intangible assets are hallmarks of a growth-by-acquisition strategy.

- ☐ Long-term debt

Incorrect Answer Feedback: Incorrect because although long-term debt may have changed (likely increased) as a result of a growth-by-acquisition strategy, it may also have changed because of changes to the target capital structure or a share-based acquisition strategy.

- ☐ Property, plant and equipment

Incorrect Answer Feedback: Incorrect because although property, plant and equipment may have changed (likely increased) as a result of a growth-by-acquisition strategy, it may also have changed because of significant new direct investments in capital. In addition, the nature of DB's industry suggests that acquisitions may include significant intangible assets (e.g., brands) that would not significantly change DB's property, plant and equipment.

13 Multiple Choice 0 / 1 point

After Winslow's adjustment, DB's net income (in EUR millions) under management's direct control is:

- ☒ 793.

Incorrect Answer Feedback: Incorrect because it correctly excludes income from investments in associates but incorrectly excludes net income attributable to non-controlling interests.

Net income attributable to shareholders	984
Less: Income from investments in associates	(191)
	793

Including the net investments and returns of associates with the full reported value of Nestlé's own assets and income would introduce noise into the analytical signals. The returns earned by associates are not under the direct control of Nestlé's management.

- ☐ 879.

Correct Answer Feedback: Correct because it excludes the income from investments in associates which is not under DB's direct control (in EUR millions):

Net income	1,070
Less: Income from investments in associates	(191)
	879

Including the net investments and returns of associates with the full reported value of Nestlé's own assets and income would introduce noise into the analytical signals. The returns earned by associates are not under the direct control of Nestlé's management.

- ☐ 984.

Incorrect Answer Feedback: Incorrect because this answer is net income attributable to shareholders which incorrectly includes income from investments in associates but incorrectly excludes net income attributable to non-controlling interests:

Net income	1,070
Less: Net income attributable to non-controlling interests	(86)
Net income attributable to shareholders	984

Including the net investments and returns of associates with the full reported value of Nestlé's own assets and income would introduce noise into the analytical signals. The returns earned by associates are not under the direct control of Nestlé's management.

14 Multiple Choice 1 / 1 point

Which of the following reasons *best* supports Winslow's disagreement with her team about DB's overall performance?

- ☐ Reason 1

Incorrect Answer Feedback: Incorrect because a pattern of steady cash-flow-based accruals is unlikely to raise concerns. An erratic or upward-moving trend in cash-flow-based accruals would be concerning. The analyst would have been more concerned if the absolute levels of the accruals ratio were high; even more worrisome would have been if they were consistently trending higher.

☐ Reason 2

Incorrect Answer Feedback: Incorrect because it is upward-trending relative accruals that could suggest management is trying to manipulate reported performance. Downward trending accruals provide less opportunity for manipulation. The analyst notes that the absolute level of accruals on the balance sheet is minor relative to the size of the operating assets, on either an ending balance basis or an average basis. The analyst would have been more concerned if the absolute levels of the accruals ratio were high; even more worrisome would have been if they were consistently trending higher.

☒ Reason 3

Correct Answer Feedback: Correct because the quality of income from operations is considered better when it is backed by cash flow. In this case, cash generated from operations is not fully backing income from operations, thereby calling the quality of income from operations into question.

If the cash flow generated by operations was significantly or consistently less than operating profit, one would have reason to be suspicious about the quality of the operating profit.

15 Multiple Choice 1 / 1 point

With respect to the segment analysis, which of the following statements is correct?

☐ Statement 1

Incorrect Answer Feedback: Incorrect because per the table below (in EUR millions except where indicated), the sales growth rate in carbonated drinks - North America is lower, and thus weaker, than that of Energy Drinks, not stronger.

	Carbonated drinks - North America	Energy Drinks	Analysis
Sales - Most recent year (A)	1,913	284	---
Sales - Previous year (B)	1,721	202	---
Sales growth % (A/B)	11.2%	40.6%	Carbonated drinks - North America lower

☒ Statement 2

Correct Answer Feedback: Correct because per the table below (in EUR millions except where indicated), carbonated drinks - Europe's EBIT margin is higher than, and thus superior to that of carbonated drinks - North America.

	Carbonated drinks - Europe	Carbonated drinks - North America	Analysis
EBIT (A)	955	558	---
Sales (B)	2,845	1,913	---
EBIT margin % (A/B)	33.6%	29.2%	Carbonated drinks - Europe higher

☐ Statement 3

Incorrect Answer Feedback: Incorrect because per the table below (in EUR millions except where indicated), carbonated drinks - Europe's asset turnover is lower, and thus worse, than that of DB overall, not better.

	Carbonated drinks - Europe	DB overall	Analysis
Sales (A)	2,845	5,577	---
Assets (B)	12,416	22,172	---
Asset turnover (A/B)	0.23x	0.25x	Carbonated drinks - Europe lower

Vignette

Claus Petersen, a pension fund equity analyst, is preparing an analysis of Rhine AG for the upcoming quarterly fund meeting. Rhine is a Germany-based manufacturer that operates three distinct divisions: children's products (infant car seats, strollers, cribs, etc.), recreational products (bicycles, bicycle trailers, etc.), and home furnishings (contemporary furniture). All three divisions sell through retail outlets around the world.

The company has been pursuing an aggressive growth strategy, achieved through both foreign acquisitions and organic growth. Petersen is interested in determining how well Rhine is allocating its resources between the three divisions and the effects of the foreign acquisitions on overall performance. Exhibit 1 summarizes selected divisional and corporate data for the current year and the prior year.

Exhibit 1 Rhine AG Selected Divisional and Corporate Data (€ millions)

	Total for Three Divisions		Children's Products		Recreational Products		Home Furnishings	
	Current Year	Prior Year	Current Year	Prior Year	Current Year	Prior Year	Current Year	Prior Year
Revenues	2,837.1	2,775.5	1,176.2	1,236.2	1,034.1	930.0	626.8	609.3
Gross profit	621.4	640.8	296.6	337.6	246.0	220.3	78.8	82.9
Operating profit	172.7	219.4	64.7	115.7	72.9	62.2	35.1	41.5
Earnings before taxes (EBT)	136.6	170.0	---	---	---	---	---	---
Net earnings after tax	109.9	132.3	---	---	---	---	---	---
Total assets	2,498.0	2,479.5	1,270.9	1,249.6	961.5	948.5	265.6	281.4
Capital expenditures	32.7	42.3	22.1	30.0	6.7	8.6	3.9	3.7
Proportion of capital expenditures	100%	100%	67.6%	70.9%	20.5%	20.3%	11.9%	8.7%
Proportion of total assets	100%	100%	50.9%	50.4%	38.5%	38.3%	10.6%	11.3%

Petersen's preferred method to determine which division is becoming less significant over time is to review the relationship between capital expenditures and total assets by operating division. He plans to base his conclusion on the assumption that the current year's investment behavior is representative of future investment patterns.

Petersen knows that revenues in the children's products division have suffered because of declining birth rates in Europe and North America, but he believes that if Rhine can maintain the operating margin for this division then overall company profitability should not be affected.

Corinna Berg, another analyst with the fund, reminds Petersen that during the current year, the US dollar weakened against the euro by 4% and that 50% of the sales in the recreational products division are sold in the United States.

Petersen recalls that some of the recent global expansion was aimed at establishing operations in Ireland because its statutory corporate tax rate is lower than the German rate of 29.8%. If Petersen assumes that other tax credits were the same in the current year as the prior year, he can analyze changes in Rhine's effective tax rate to determine whether the geographic mix of the company's profits has changed in the current year.

Petersen finally examines the company's liquidity ratios, which are shown in Exhibit 2. Even though the company's current and quick ratio have improved, his interpretation of the changes in the company's cash conversion cycle is that the company's liquidity position has deteriorated.

Exhibit 2 Rhine AG Selected Ratios

Ratio	Current Year	Prior Year	2 Years Prior
Current ratio	2.31	2.17	1.16
Quick ratio	1.06	0.89	0.53
Accounts receivable turnover	5.82	6.08	6.11
Inventory turnover	3.78	3.91	4.09
Accounts payable turnover	5.71	5.78	5.60
Cash conversion cycle	95 days	90 days	84 days

Worried that the balance sheet–based and cash flow–based accruals ratios (not shown) raise some concerns about the possible use of accruals to manage earnings, Petersen asks Berg for advice on what further type of analysis he should do as a follow-up on this issue.

16 Multiple Choice 1 / 1 point

Using Petersen's preferred method and current year divisional data, the *best* conclusion Peterson can make about which division will potentially become less significant in the future is that it will be:

- ☐ home furnishings.

Incorrect Answer Feedback: Incorrect. Petersen prefers to use the relationship between capital expenditures and total assets by operating division and thus would use the ratio of capital expenditure proportion to total asset proportion for each division. This ratio for the recreational products division is less than 1 (see table), indicating that Rhine is allocating a lower proportion of capital expenditures to that division relative to asset proportions. If this trend continues, the recreational products division will become less significant over time.

	Children's Products	Recreational Products	Home Furnishings	Total for the Three Divisions
Proportion of capital expenditures (Exhibit 1)	67.6%	20.5%	11.9%	100.0%
Proportion of total assets (Exhibit 1)	50.9%	38.5%	10.6%	100.0%
Ratio: Proportion of capital expenditures to proportion of total assets	1.3	0.5	1.1	1.0

- ☐ children's products.

Incorrect Answer Feedback: Incorrect. Petersen prefers to use the relationship between capital expenditures and total assets by operating division and thus would use the ratio of capital expenditure proportion to total asset proportion for each division. This ratio for the recreational products division is less than 1 (see table), indicating that Rhine is allocating a lower proportion of capital expenditures to that division relative to asset proportions. If this trend continues, the recreational products division will become less significant over time.

	Children's Products	Recreational Products	Home Furnishings	Total for the Three Divisions
Proportion of capital expenditures (Exhibit 1)	67.6%	20.5%	11.9%	100.0%
Proportion of total assets (Exhibit 1)	50.9%	38.5%	10.6%	100.0%
Ratio: Proportion of capital expenditures to proportion of total assets	1.3	0.5	1.1	1.0

☒ recreational products.

Correct Answer Feedback: Correct. Petersen prefers to use the relationship between capital expenditures and total assets by operating division and thus would use the ratio of capital expenditure proportion to total asset proportion for each division. This ratio for the recreational products division is less than 1 (see table), indicating that Rhine is allocating a lower proportion of capital expenditures to that division relative to asset proportions. If this trend continues, the recreational products division will become less significant over time.

	Children's Products	Recreational Products	Home Furnishings	Total for the Three Divisions
Proportion of capital expenditures (Exhibit 1)	67.6%	20.5%	11.9%	100.0%
Proportion of total assets (Exhibit 1)	50.9%	38.5%	10.6%	100.0%
Ratio: Proportion of capital expenditures to proportion of total assets	1.3	0.5	1.1	1.0

17 Multiple Choice 1 / 1 point

If the children's products division had been able to maintain its prior year operating margin in the current year, the company's overall operating margin in current year, compared to the prior year, would have been:

☒ lower.

Correct Answer Feedback: Correct. Apply the prior year operating margin for the children's products division to the current year revenues for the division to determine what the current year overall operating profit margin would have been if the margin had been maintained, and then compare it with the prior year overall operating profit margin.

Operating margin in children's products in prior year = Operating profit/revenues	115.7/1,236.2	9.4%
Apply the 9.4% margin to current year revenues for division	$0.094 \times 1,176.2$	110.6
Increase in current year operating profit for division, had the margin been maintained	$110.6 - 64.7$	45.9
Revised current year company operating profit with increase in children's products operating profit	$172.7 + 45.9$	218.6
Revised current year company operating margin	$218.6/2,837.1$	7.7%
Prior year company operating margin	$219.4/2,775.5$	7.9%

Even if the children's products division had maintained its operating margin in the current year, the overall company operating margin would still have decreased slightly (7.7% versus 7.9%).

☐ the same.

Incorrect Answer Feedback: Incorrect. It is not the same.

☐ higher.

Incorrect Answer Feedback: Incorrect: If they incorrectly add the full 110.6 to the operating profit (and not just the incremental change) then it would be higher: $(172.7 + 110.6)/2,837.1 = 10\%$.

18 Multiple Choice 1 / 1 point

Compared with 2 years prior, the change in which working capital account *most likely* had the largest effect on Petersen's observed deterioration in liquidity?

☒ Inventory

Correct Answer Feedback: Correct. Petersen interprets that the changes in the cash conversion cycle (CCC) indicate a deterioration in liquidity. The CCC has increased since 2 years prior, from 84 days to 95 days (see the following table). The working capital account that had the largest effect on the increase was inventory because the holding period has increased 6.4 days.

	Current Year		Prior Year	
		Days		
Working Capital Account	Turnover	(365/Turnover)	Turnover	Days
Accounts receivable	5.82	62.7	6.11	59.8
Inventory	3.78	96.6	4.09	89.2
Accounts payable	5.71	63.9	5.60	65.2
Cash conversion cycle*		95.4		83.8

* CCC = Days in sales + Days in inventory – Days in payables.

☐ Accounts payable

Incorrect Answer Feedback: Incorrect. Accounts payable management has stayed fairly constant. In fact, they are paying off their A/P a little faster than before, which would be a sign of improved liquidity.

☐ Accounts receivable

Incorrect Answer Feedback: Incorrect. Accounts receivable management did deteriorate with collections slowing by three days, but the change in inventory management was greater at approximately six days.

19 Multiple Choice 1 / 1 point

Berg's best answer to Petersen's question about further analysis is that he should conduct a:

☐ DuPont analysis.

Incorrect Answer Feedback: Incorrect. Concerns about earnings manipulation are best addressed by cash flow ratios, such as operating cash flow before interest and taxes to operating income.

DuPont provides useful information about the components of a company's ROE but would not detect earnings management.

☒ Cash flow ratio analysis.

Correct Answer Feedback: Correct. Concerns about earnings manipulation are best addressed by cash flow ratios, such as operating cash flow before interest and taxes to operating income.

☐ Discounted cash flow analysis.

Incorrect Answer Feedback: Incorrect. Concerns about earnings manipulation are best addressed by cash flow ratios, such as operating cash flow before interest and taxes to operating income.

Cash flow projections would be more useful for determining valuation